Bocconi

CORPORATE FINANCE 30017

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Equity Valuation

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Agenda

EQUITY MARKET INSTITUTIONS

DIVIDEND DISCOUNT MODEL

PAYOUT POLICY

REMARKS ON VALUATION

EQUITY MARKET INSTITUTIONS

Equity contracts

- In general less complex than debt contracts (why, possibly?)

- Right to receive **dividend payments** (at discretion of company directors)
- Right to **vote in shareholder meeting** (optional)
  - Recall *Snap Inc.*
  - "Dual class shares"/"non-voting shares"
  - "Preferred shares" contractually rank senior to common shares (dividend, liquidation cash flows).

American Depository Receipts (ADRs)

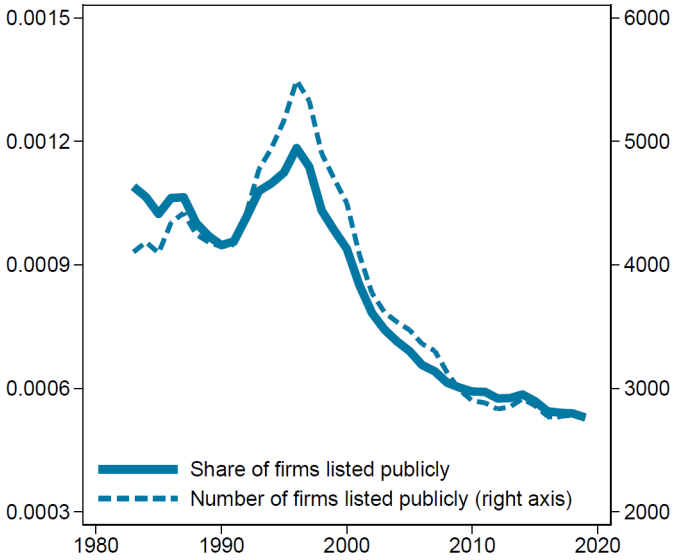
- Quasi shares in foreign companies (US *cross-listing* of non-US firms)
  - Issued by US depositary bank, represents (fraction or multiple) of shares of foreign stock
  - Quoted in USD, pay dividends in USD, and can be traded like the shares of US-based companies.
- Way of getting around (some) US listing regulations
- Price close to the price of the foreign stock in its home market (up to arbitrage limits)

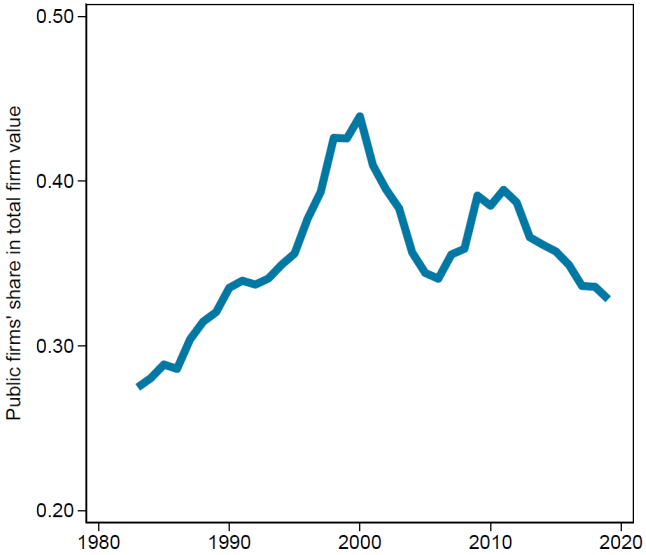
Equity Market Institutions

- Primary markets: Sales of new shares to raise new capital
- Secondary markets: Trading of secondhand shares (no new capital raised)

- Stock markets can be organized in different ways:
  - Exchanges: *centralized*
    - Auction markets (many details, technology matters)
      - NYSE, Tokyo Stock Exchange, London Stock Exchange, Frankfurt
      - Specialist acts as auctioneer. Typically a computer nowadays.
  - Over-the-counter: *decentralized* (rare for equities relative to bonds)
    - Dealer markets
      - NASDAQ
      - All trades take place between investor and dealer

Minority of US non-fin. firms listed publicly





DIVIDEND DISCOUNT MODEL

Dividend Discount Model (DDM)

— Dividend Discount Model = Valuing Common Stocks via Present Value

- In words: *Stock's value equals sum of discounted expected dividends*
- **Uncertainty** of future cash flows (dividends) is especially important!
- General formulation (via stochastic discount factor $m_{h,s}$ for state s at horizon h):

$$PV = \sum_{h=1}^{\infty} \boldsymbol{E}[m_h d_h]$$

- Using discount rate $\mathrm{r} = \left(\frac{E[d_h](1+r_f)^h}{E[d_h]+(1+r_f)^h \boldsymbol{Cov(m_h,d_h)}} \right)^{\frac{1}{h}} - 1$, can rewrite into familar form:

$$PV = \sum_{h=1}^{\infty} \frac{\boldsymbol{E}[d_h]}{(1+r)^h}$$

- *Check what happens to the discount rate if* $Cov(m_h, d_h) = 0$ *(and what does it mean?)*

Basic DDM Acrobatics

— Equating price and present value and rearranging the baseline DDM equation yields…

$$P_0 = \frac{\boldsymbol{E}[d_1] + P_1}{(1 + r)}$$

- Further rearranging yields: Return = Dividend Yield + Capital Gain

$$r = \frac{\boldsymbol{E}[d_h]}{P_0} + \frac{P_1 - P_0}{P_0}$$

Special Case: Constant Dividend Growth

- Assume dividend growth to be constant:

$$\boldsymbol{E}[d_h] = \boldsymbol{E}[d_1(1+g)^h] = \boldsymbol{E}[d_1](1+g)^h$$

- Then, the DDM equation can be re-written into

$$PV = \frac{\boldsymbol{E}[d_1]}{r-g}$$

- Useful: rearranging for discount rate: $\mathrm{r} = \frac{\boldsymbol{E}[d_1]}{PV} + g$

Mini Case: Northwest Natural Gas

- Northwest Natural Gas (now known as NW Natural) is a real U.S. publicly traded utility based in Portland, Oregon, that primarily distributes natural gas to customers in the Pacific Northwest
  - Stock is selling at $41.67 per share
  - Dividend payments are expected to be $1.50.

- Utilities are mature and stable industries, ideally suited for the constant-growth model.
  - Analysts forecast annual (dividend) growth of 5.2%

- What is NW Natural's equity discount rate (aka. cost of equity)?

PAYOUT POLICY

Should we pay dividends or reinvest?

- If a firm pays lower dividend to reinvest funds, stock price may (not) increase due to higher future dividends
  - Key: *Is present value of dividend increase worth more than today's payout?*

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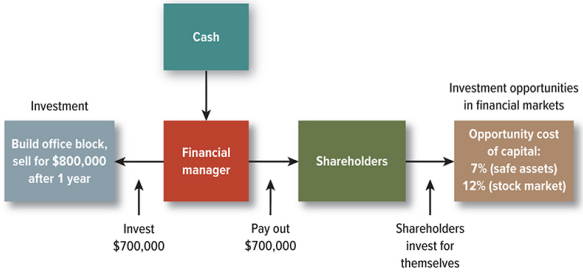

- *Recall: Invest if NPV > 0*
  - *Will increase the stock price*

- There is another way, based on a few simple balance sheet metrics (using some assumptions)…

Simple but powerful payout arithmetic

- Define:
  - "Plowback ratio": Share of earnings reinvested (θ)
  - "Payout ratio" = (1 - Plow-back ratio): Share of earnings paid out ($1-\theta$)
  - "Earnings per share": Earnings relative to number of shares (EPS)
  - "Return on Book Equity": Earnings relative to book equity (ROE)

- Assume constant ROE and constant plowback ratio:
  - Dividends:

$$d_h = EPS_h \times (1-\theta)$$

  - Growth (under above assumptions, book equity, earnings, EPS & dividends growth at same rate):

$$g = ROE \times \theta$$

- *Where is the trade-off with respect to the value of equity (stock price)?*

Mini Case: Northwest Natural Gas (cont.)

- NW Natural's board is considering to increase plowback permanently from 40% to 50%.
- Advise the board: **Will the new plowback policy increase today's stock price?**
  - Recall:
    - dividend growth is 5.2% (under 40% plowback)
    - expected next year's dividend is USD 1.50 (under 40% plowback)
    - Use equity discount rate computed before (i.e., assume higher plowback leaves risk unchanged)
    - Stock price is USD 41.67 (under 40% plowback)
  - Tip: Compute expected dividend and growth under new policy, then apply DDM!

Present Value of Growth Opportunities

- Definition: Difference between actual share price and present value without plowback ($\theta = 0$).

$$PVGO = P_0 - \frac{EPS_1}{r}$$

- Interpretation: Value generated by retaining earnings.

REMARKS ON VALUATION

Uncertainty in Equity Valuation

- Valuing established and profitable firms that pay dividends is comparatively easier, i.e., value estimates will reasonably precise.
  - PVGO are not very important for valuation
- Valuing young, loss-making firms, that face huge risk and uncertainty, and that may not generate positive cash flows for many years, is hard. Value estimates will be subject to substantial uncertainty and thus controversial.
  - PVGO form substantial component of stock price

Careful with constant growth DDM

- Valuation very sensitive to small changes in g
- We can do better than constant growth in the short term
  - Typically more granular info about short term dividend profile

- Better: DDM with two stages:
  1. Short-term growth
  2. Long-term growth (or price)

$$P_0 = \frac{DIV_1}{(1+r)} + \frac{DIV_2}{(1+r)^2} + \frac{DIV_3 + P_3}{(1+r)^3}$$

with

$$P_3 = \frac{DIV_4}{r-g} = \frac{1.24}{r-0.08}$$

We get

$$P_0 = \frac{0.50}{(1+r)} + \frac{0.60}{(1+r)^2} + \frac{1.15}{(1+r)^3} + \frac{1}{(1+r)^3}\frac{1.24}{r-0.08}$$

Company Valuation

- Principle of PV valuation extends to entire companies
- Relevant cash flow concept? **"Free Cash Flow" (FCF)**
  - Cash flow after tax & investments needed for projected growth, but before interest & dividends

- For example, via two stage terminal value approach:

$$\boxed{\mathrm{PV} = \frac{\mathrm{FCF}_1}{(1+r)^1} + \frac{\mathrm{FCF}_2}{(1+r)^2} + \ldots + \frac{\mathrm{FCF}_H}{(1+r)^H} + \frac{\mathrm{PV}_H}{(1+r)^H}}$$